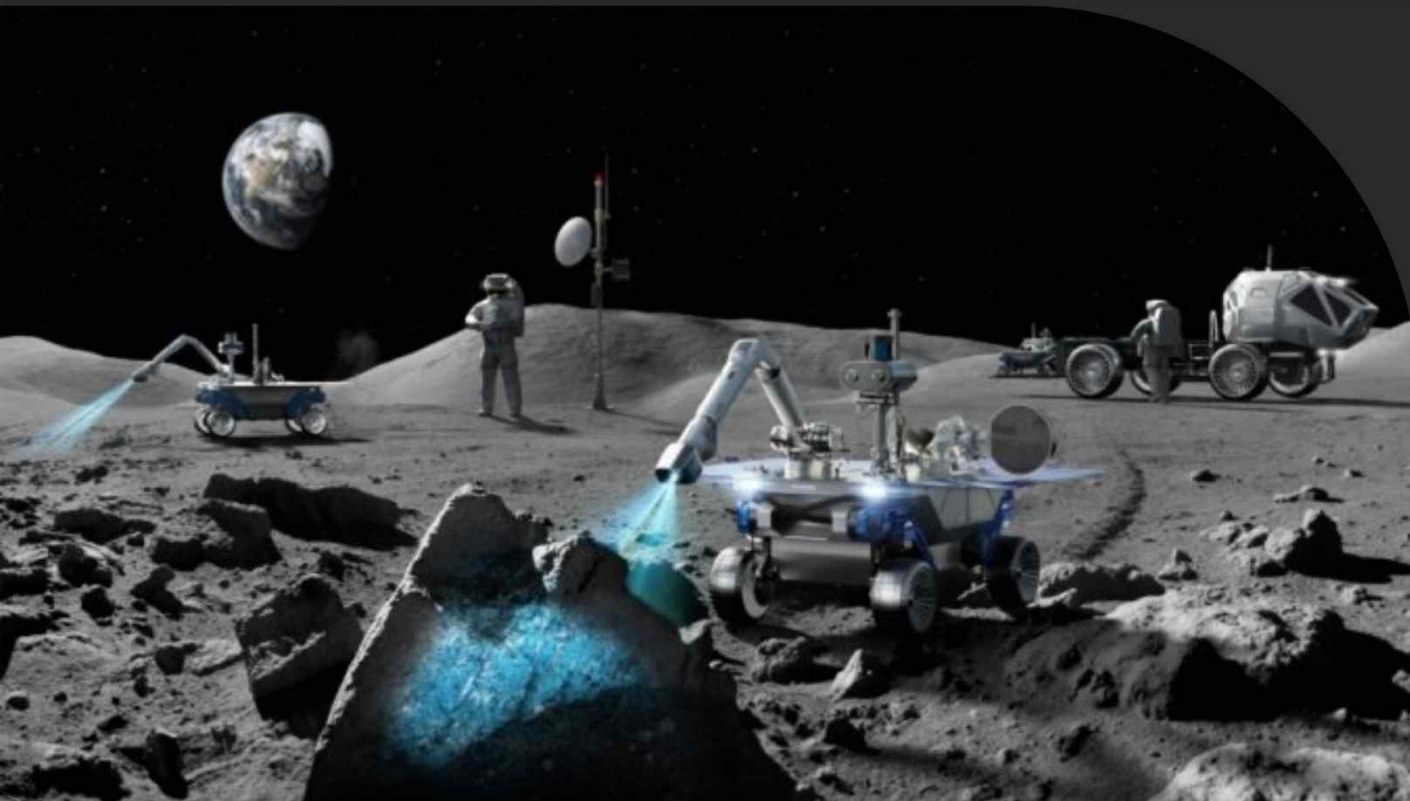


하만 커넥티드 SoC 반도체 Peripheral 설계 프로젝트



2023.08.28 ~ 2023.09.15

SoC 반도체 Peripheral 설계 프로젝트 : 오지탐사선

- 프로젝트 목표
- Pin Mapping
- FPGA 구현
- Raspberrypi 구현
- 결과

프로젝트 목표

탐사 및 데이터 수집

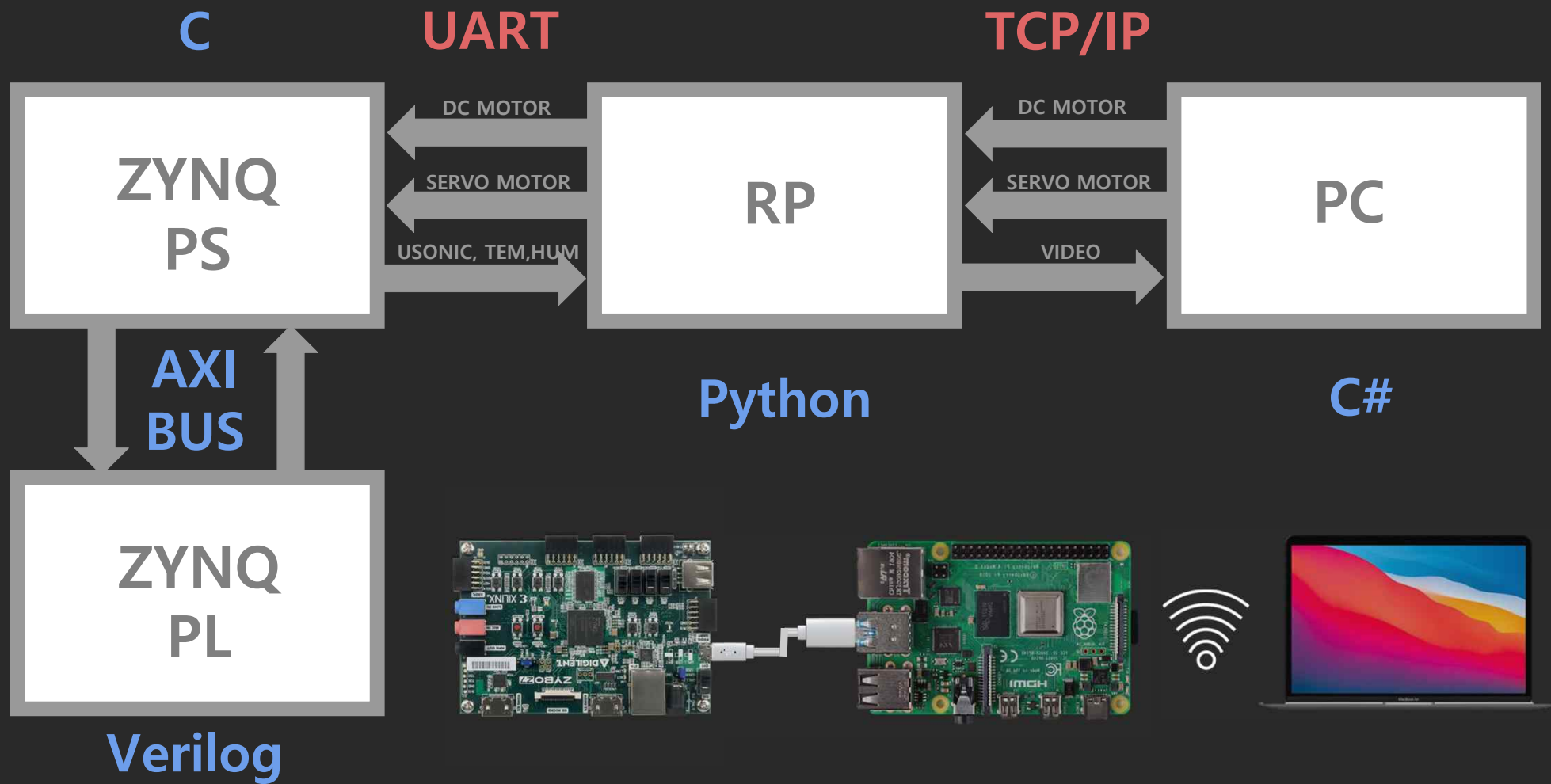
원격제어로 안전한 탐사/작업

FSM을 활용한 PWM제어

UART 및 TCP/IP를 통한 통신 인터페이스 구현

OpenCV를 이용한 이미지 처리

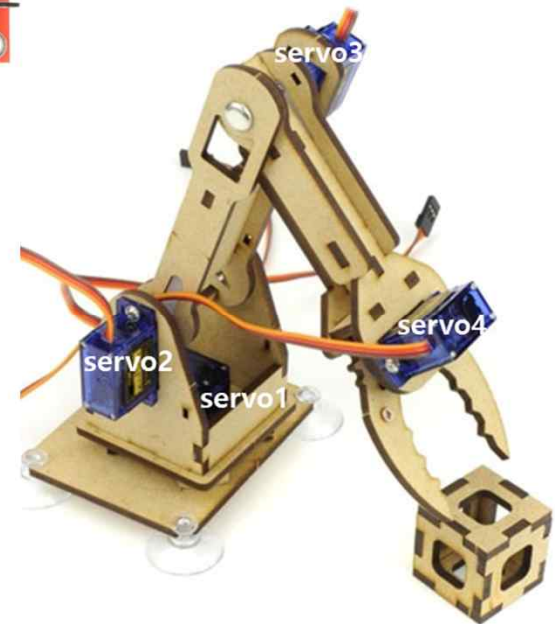
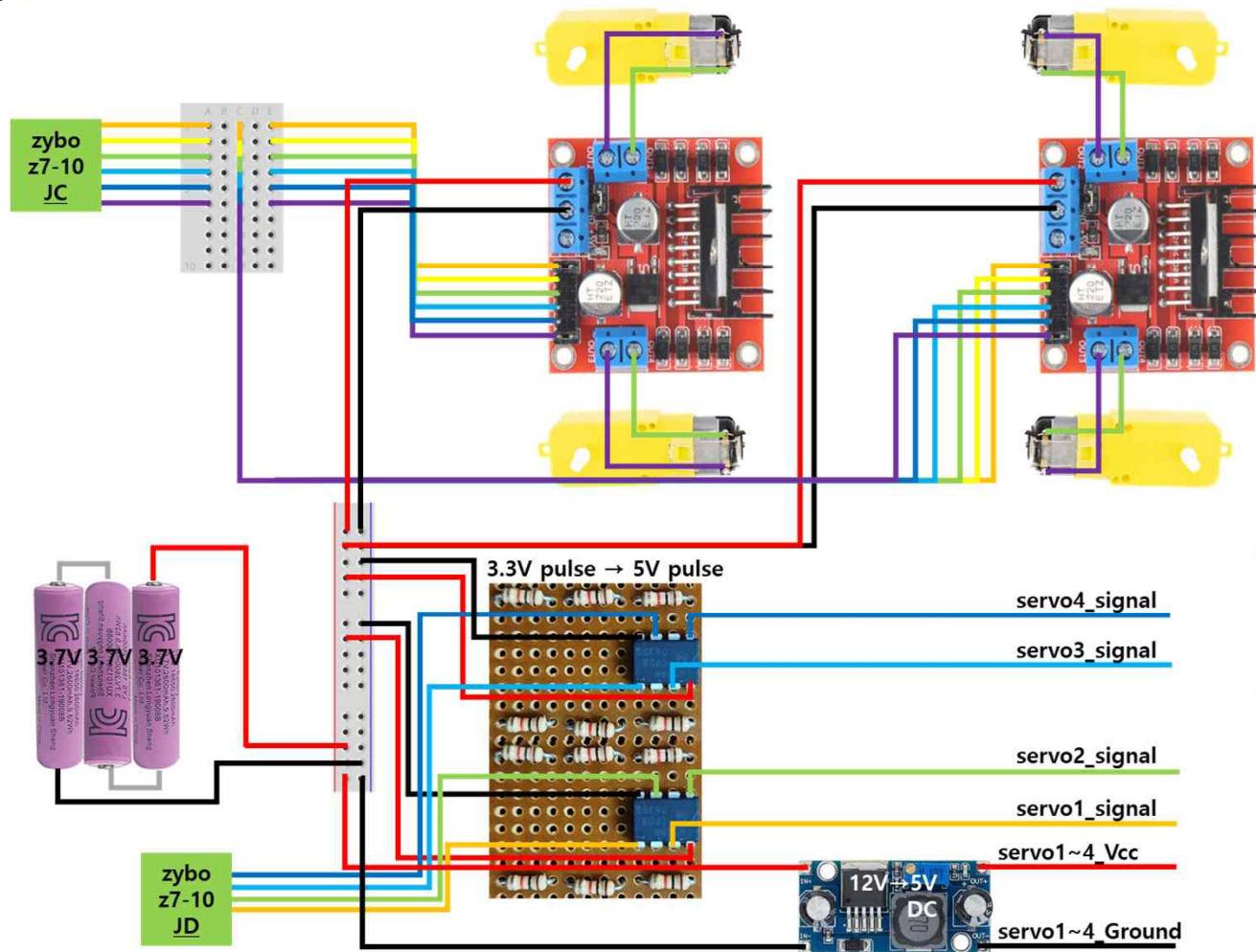
AXI BUS를 이용한 IP BLOCK설계



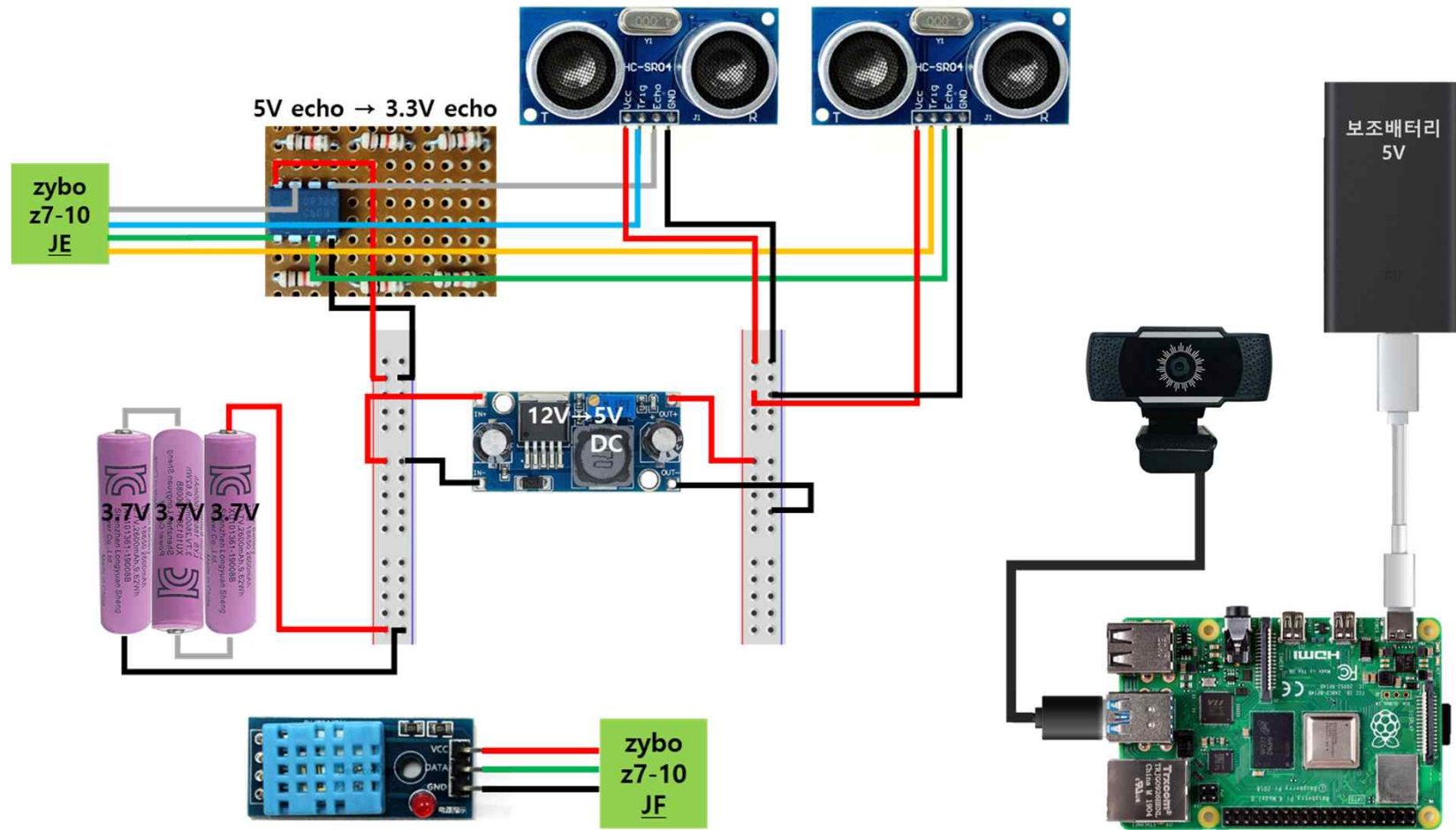
Pin Mapping



Pin Mapping



Pin Mapping

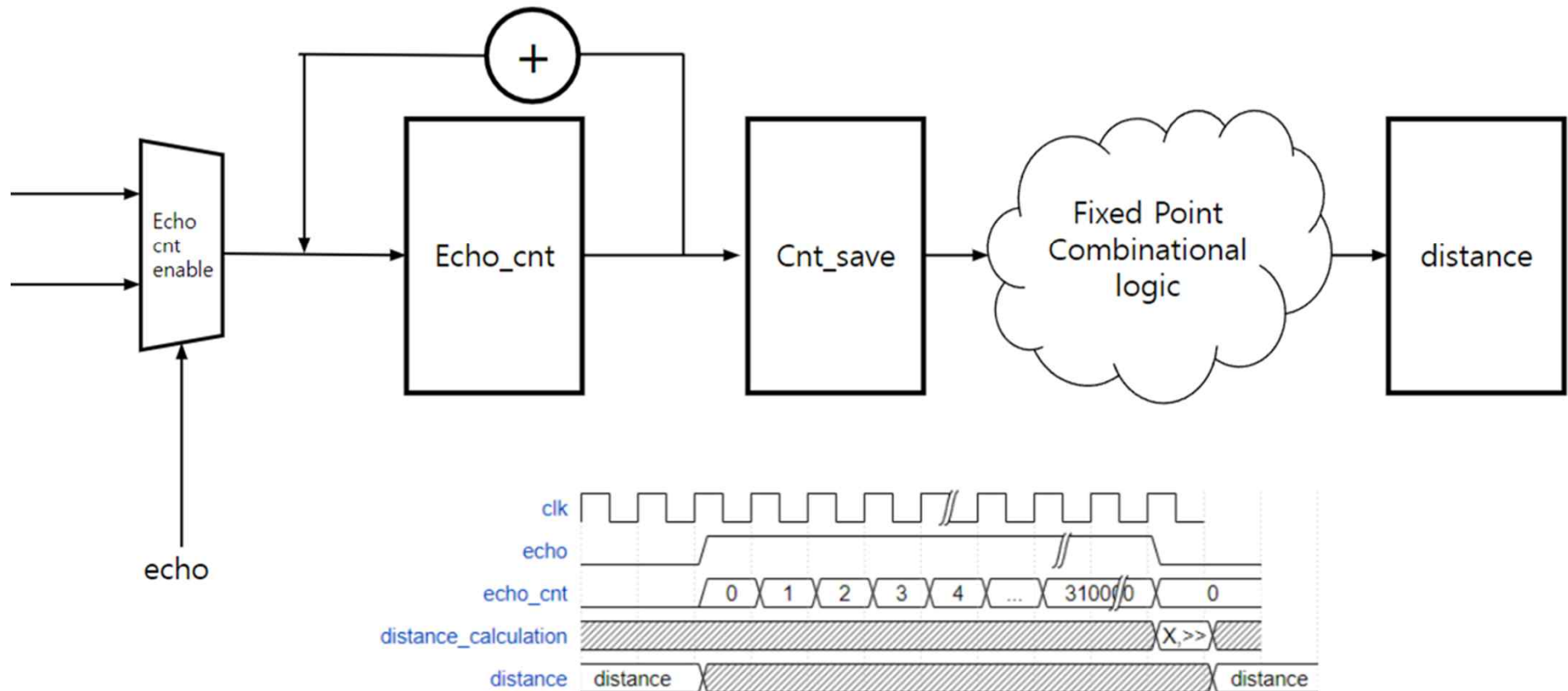


FPGA 구현

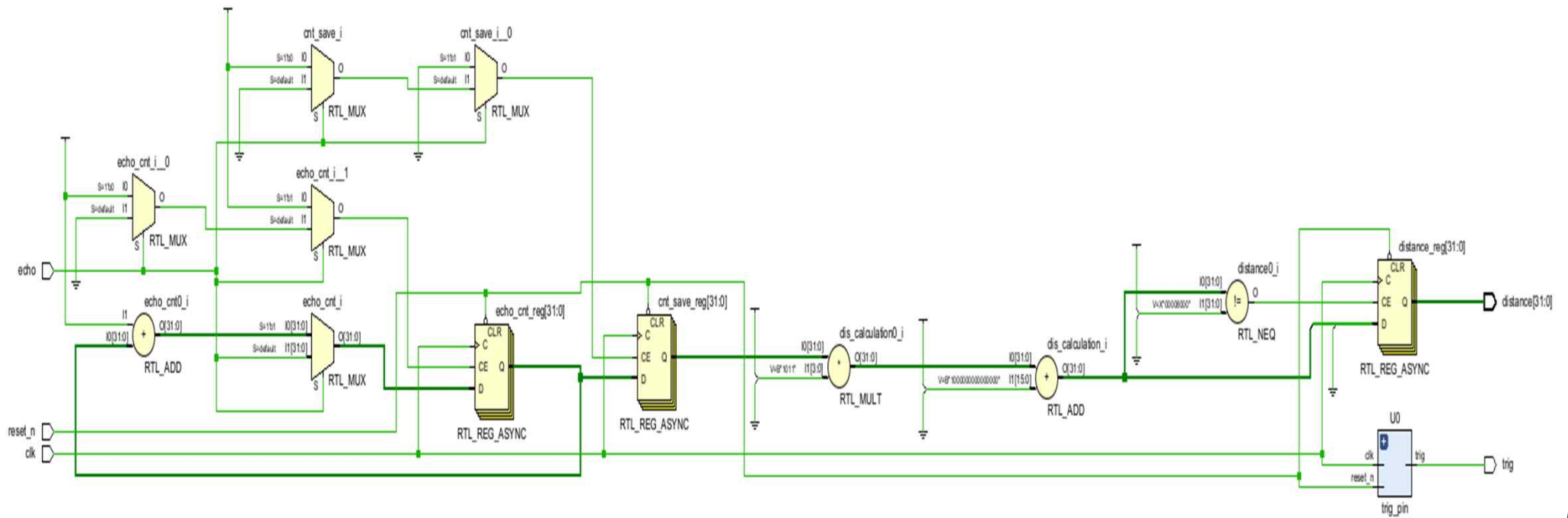
Ultra_sonic



Ultra_sonic : Block_diagram, timing diagram

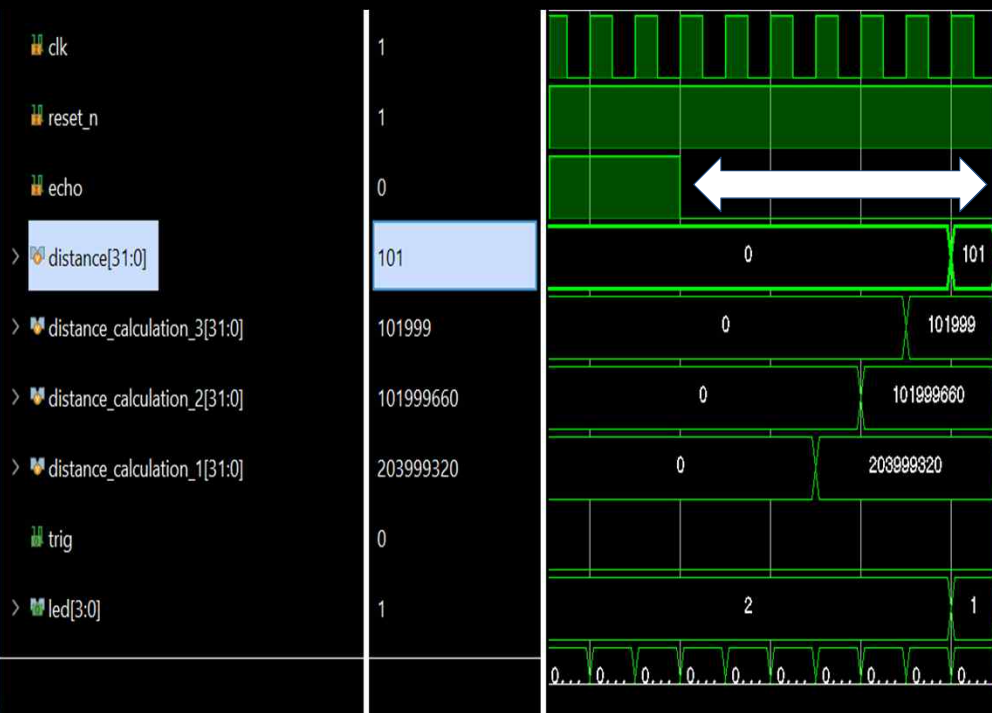


Ultra_sonic : Block_diagram



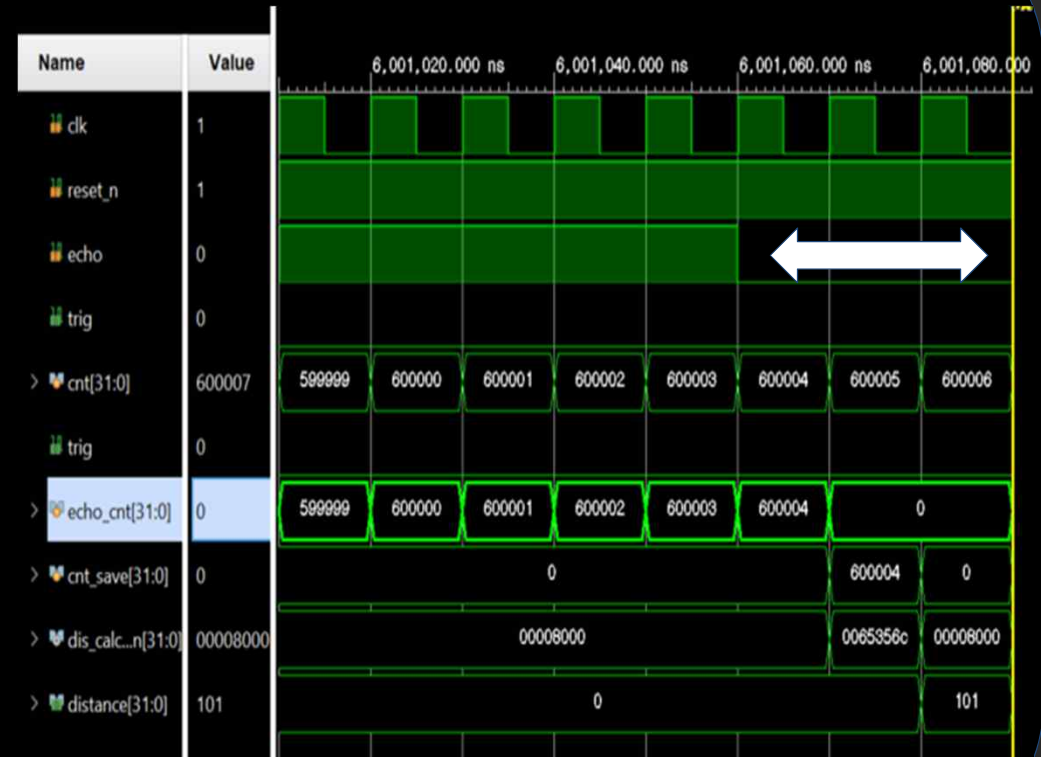
Ultra_sonic : Circuit optimization

● Divider



● Calculation Latency = 7 clock

● Fixed Point



● Calculation Latency = 3 clock

Ultra_sonic : Circuit optimization

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): -6.442 ns	Worst Hold Slack (WHS): 0.180 ns	Worst Pulse Width Slack (WPWS): 3.500 ns
Total Negative Slack (TNS): -269.758 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 44	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 290	Total Number of Endpoints: 290	Total Number of Endpoints: 231

Timing constraints are not met.



Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 0.456 ns	Worst Hold Slack (WHS): 0.157 ns	Worst Pulse Width Slack (WPWS): 3.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 126	Total Number of Endpoints: 126	Total Number of Endpoints: 112

All user specified timing constraints are met.

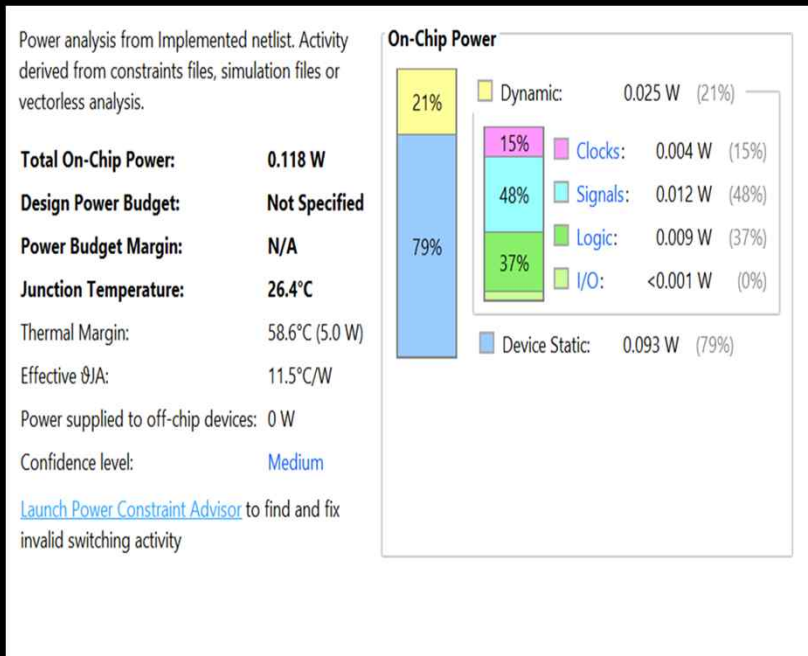
LUT	FF	BRAM
873	203	0
871	203	0



LUT 94.8% 개선
F/F 45.8% 개선

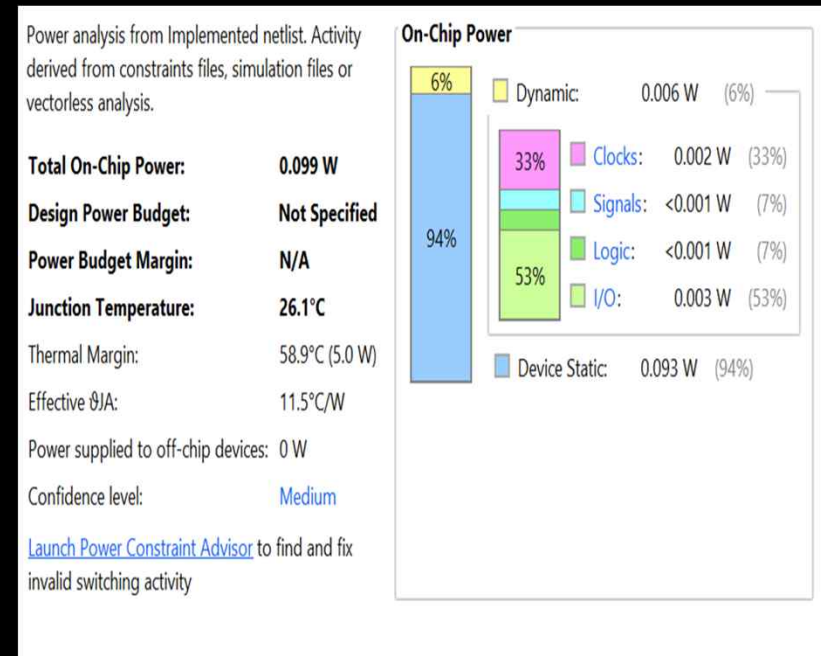
LUT	FF	BRAM
144	111	0
132	111	0

Ultra_sonic : Circuit optimization



0.118W

16.1% 개선



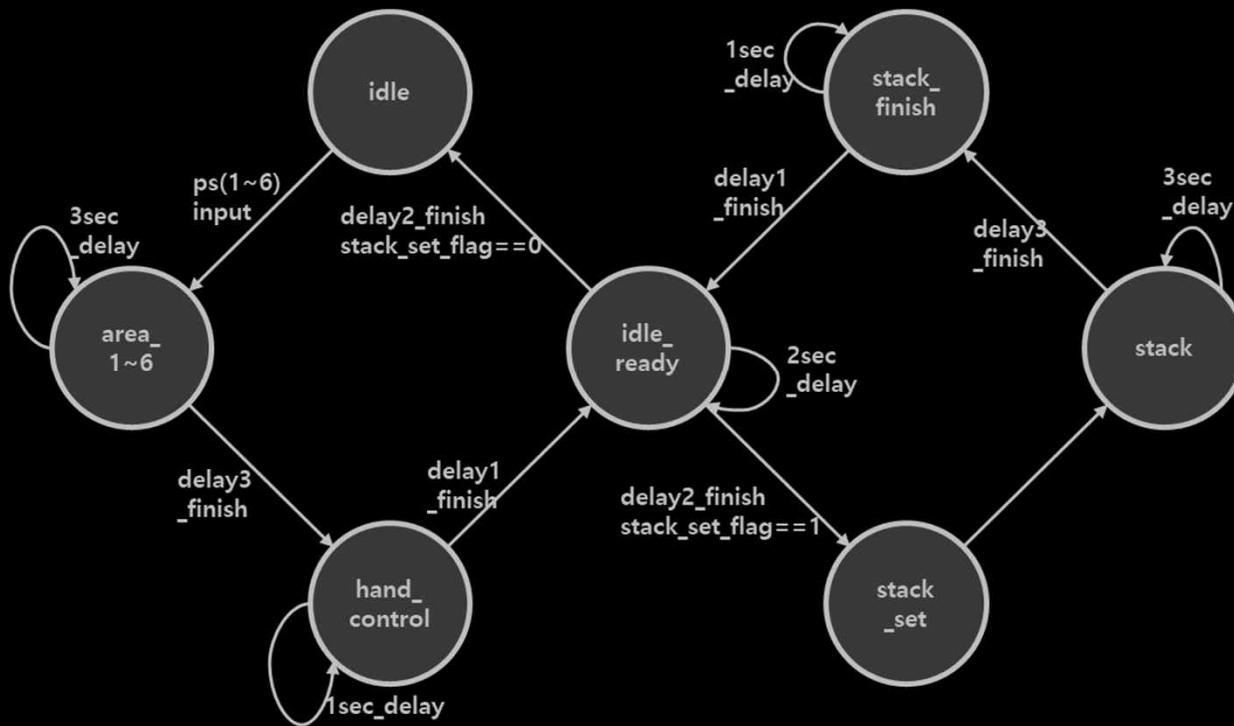
0.099W

FPGA 구현

Robot_Arm



Robot_Arm_fsm

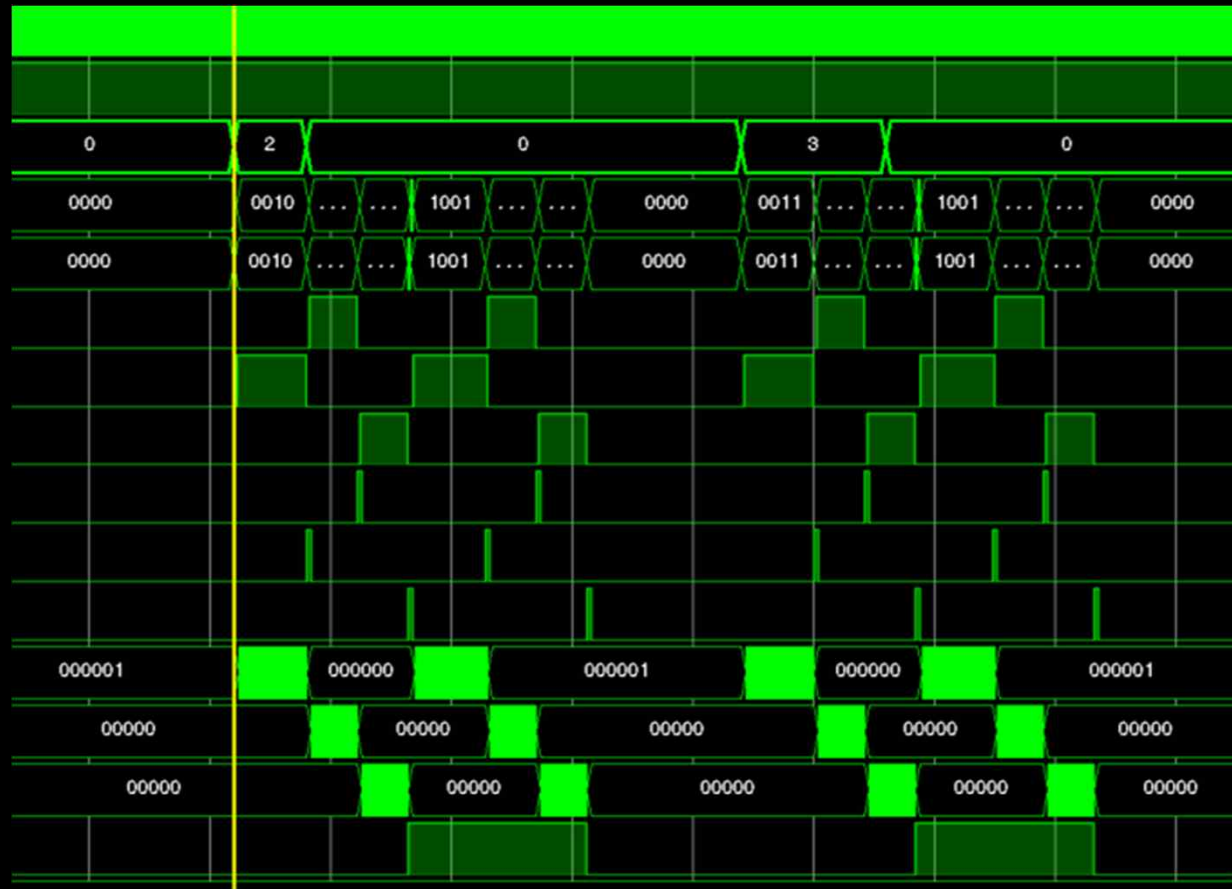


```

localparam idle           = 5'b00000;
localparam area_1         = 5'b00001;
localparam area_2         = 5'b00010;
localparam area_3         = 5'b00011;
localparam area_4         = 5'b00100;
localparam area_5         = 5'b00101;
localparam area_6         = 5'b00110;
localparam hand_control1  = 5'b00111;
localparam idle_ready     = 5'b01000;
localparam stack          = 5'b01001;
localparam stack_finish   = 5'b01010;
localparam stack_set      = 5'b01011;
//
localparam hand_control2  = 5'b01100;
localparam hand_control3  = 5'b01101;
localparam hand_control4  = 5'b01110;
localparam hand_control5  = 5'b01111;
localparam hand_control6  = 5'b10000;
  
```

	Servo1	Servo2	Servo3	Servo4
Idle	96°	96°	120°	180°
Area1	132°	0°	72°	180°
Area2	96°	12°	84°	180°
Area3	72°	0°	48°	180°
Area4	144°	36°	144°	180°
Area5	96°	36°	144°	180°
Area6	72°	36°	156°	180°
Stack	96°	144°	0°	84°

Robot_Arm_fsm : simulation



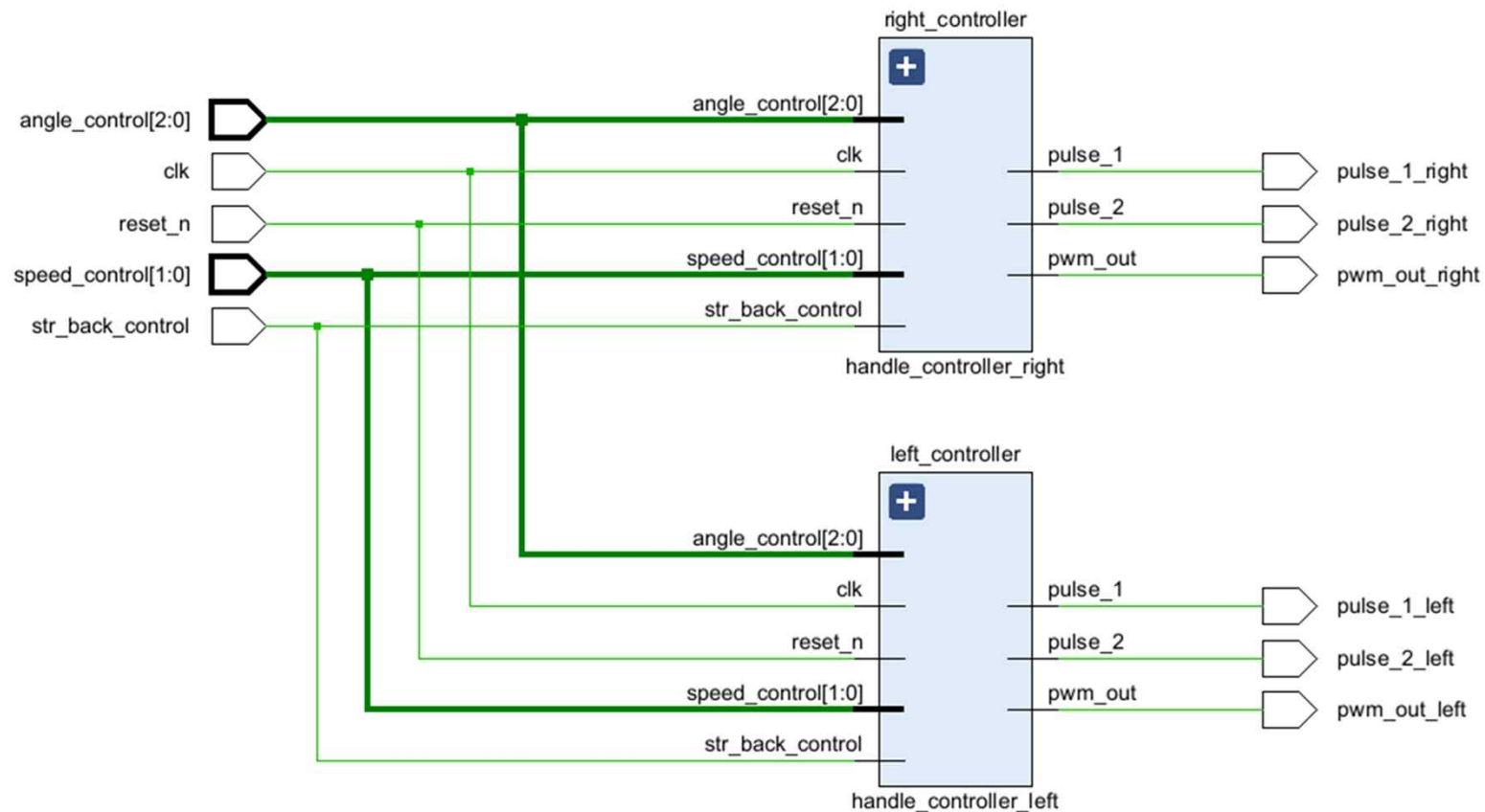
- 각 state에 따른 상태 천이 simulation

FPGA 구현

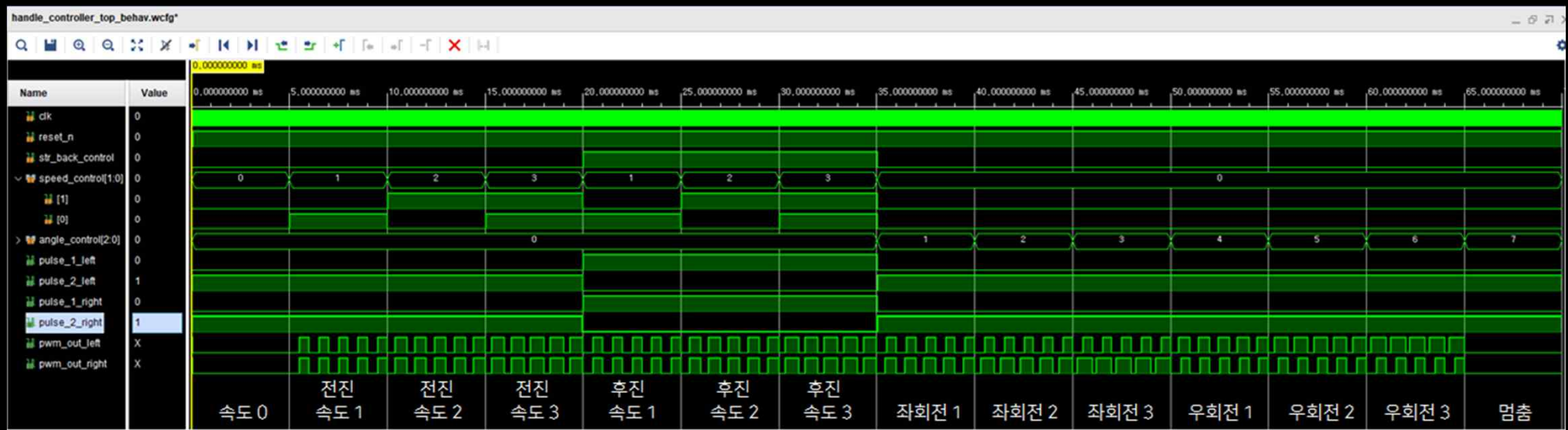
DC_motor_control



DC_motor_control : Block diagram



DC_motor_control : simulation



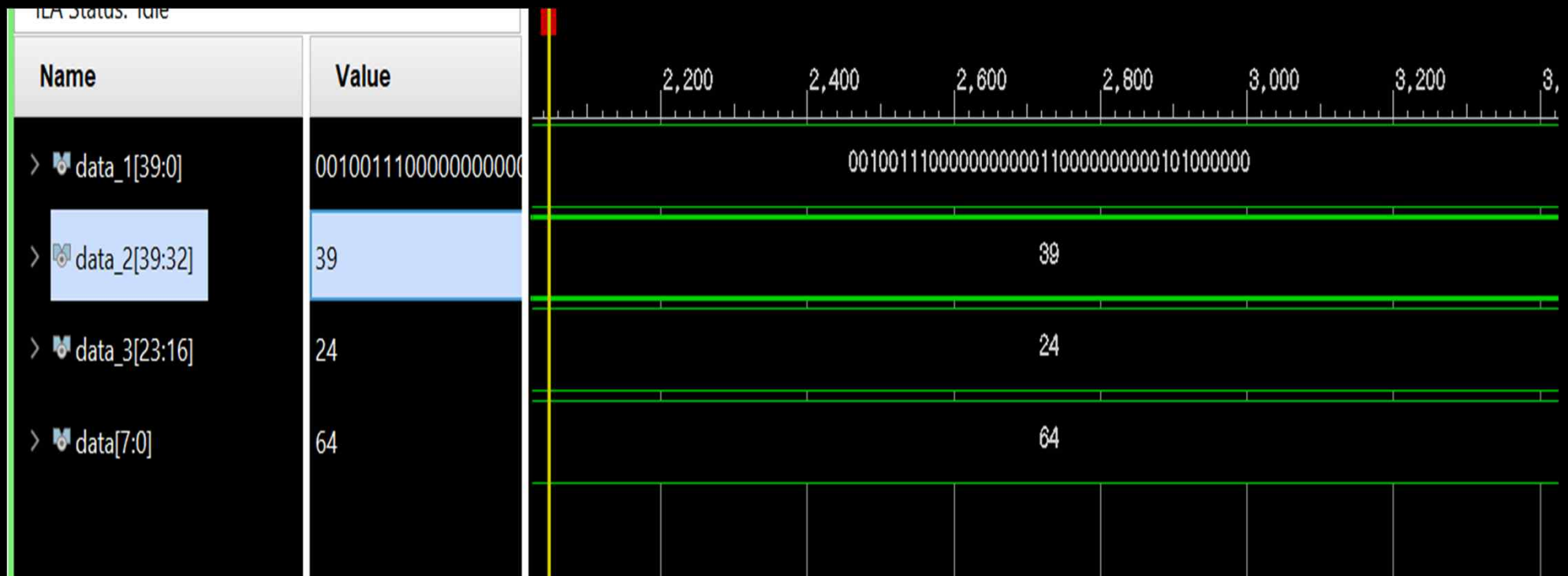
FPGA 구현

Temperature and humidity sensor



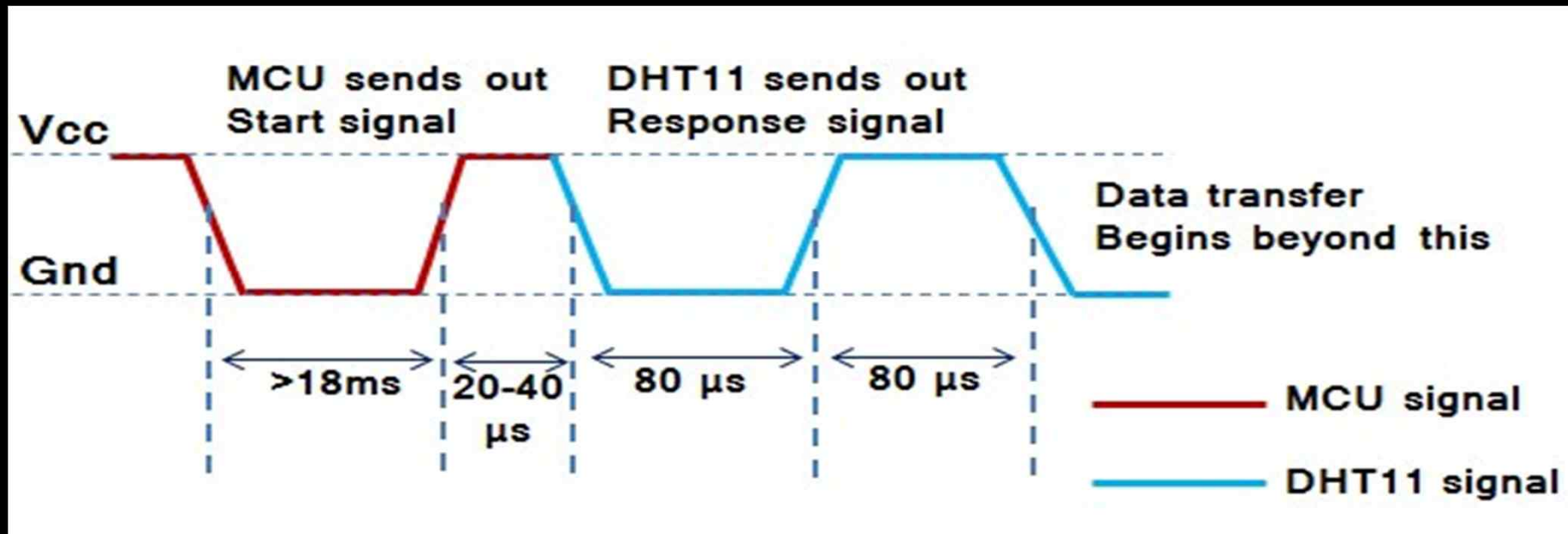
Temperature and humidity sensor

● ILA를 이용한 DHT_11 온습도 센서 data 동작 검증



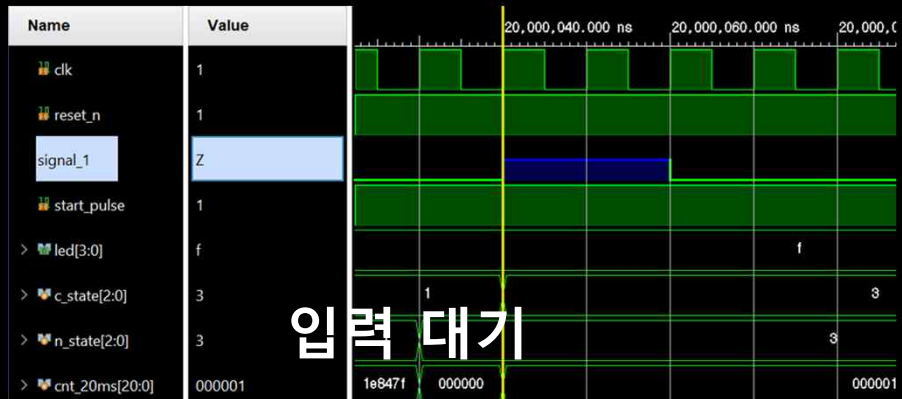
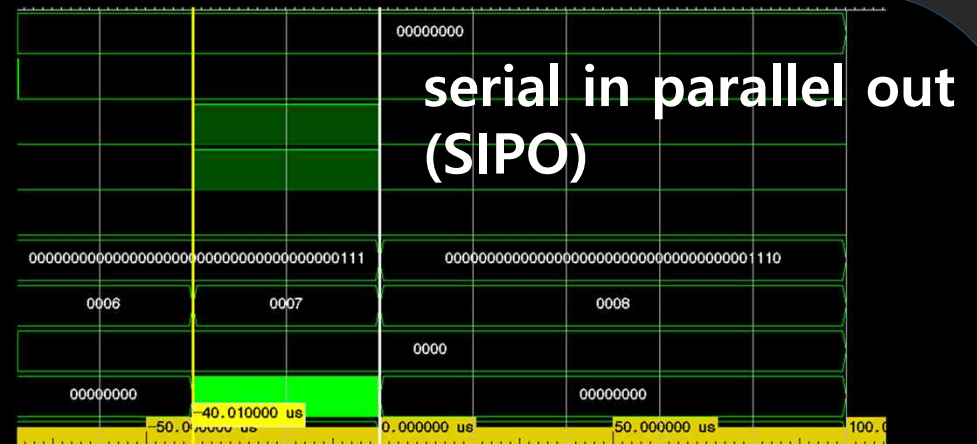
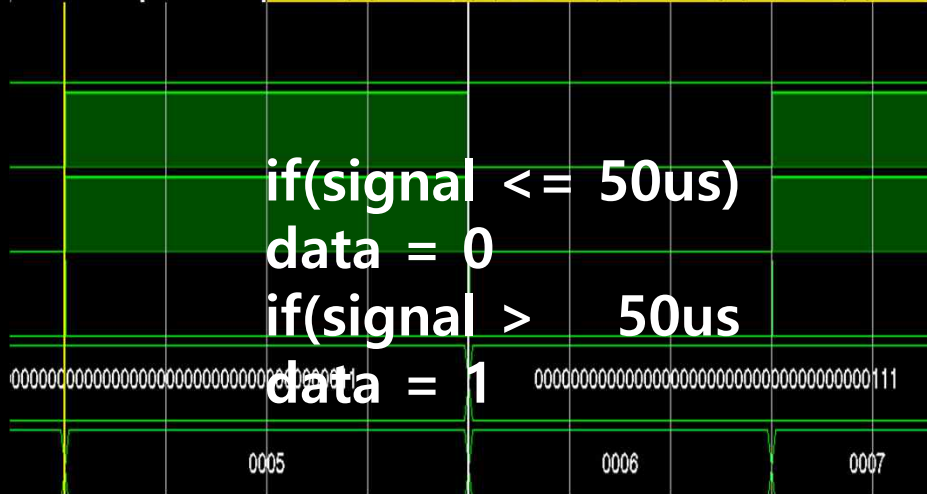
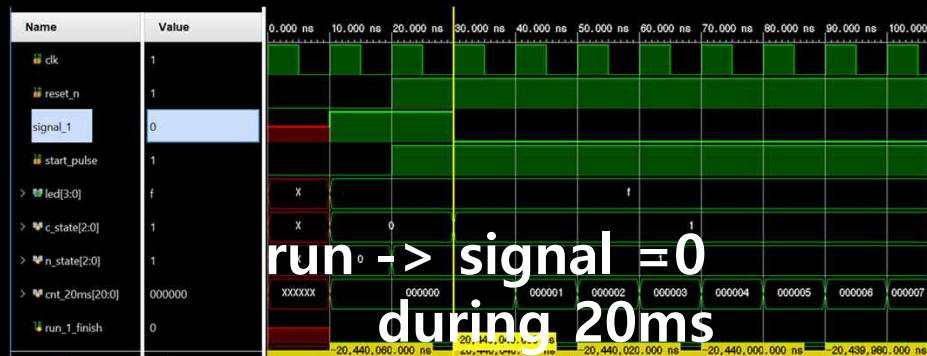
Temperature and humidity sensor

- DHT_11 온습도 센서 timing diagram



Temperature and humidity sensor

● ILA를 이용한 DHT_11 온습도 센서 data 동작 검증

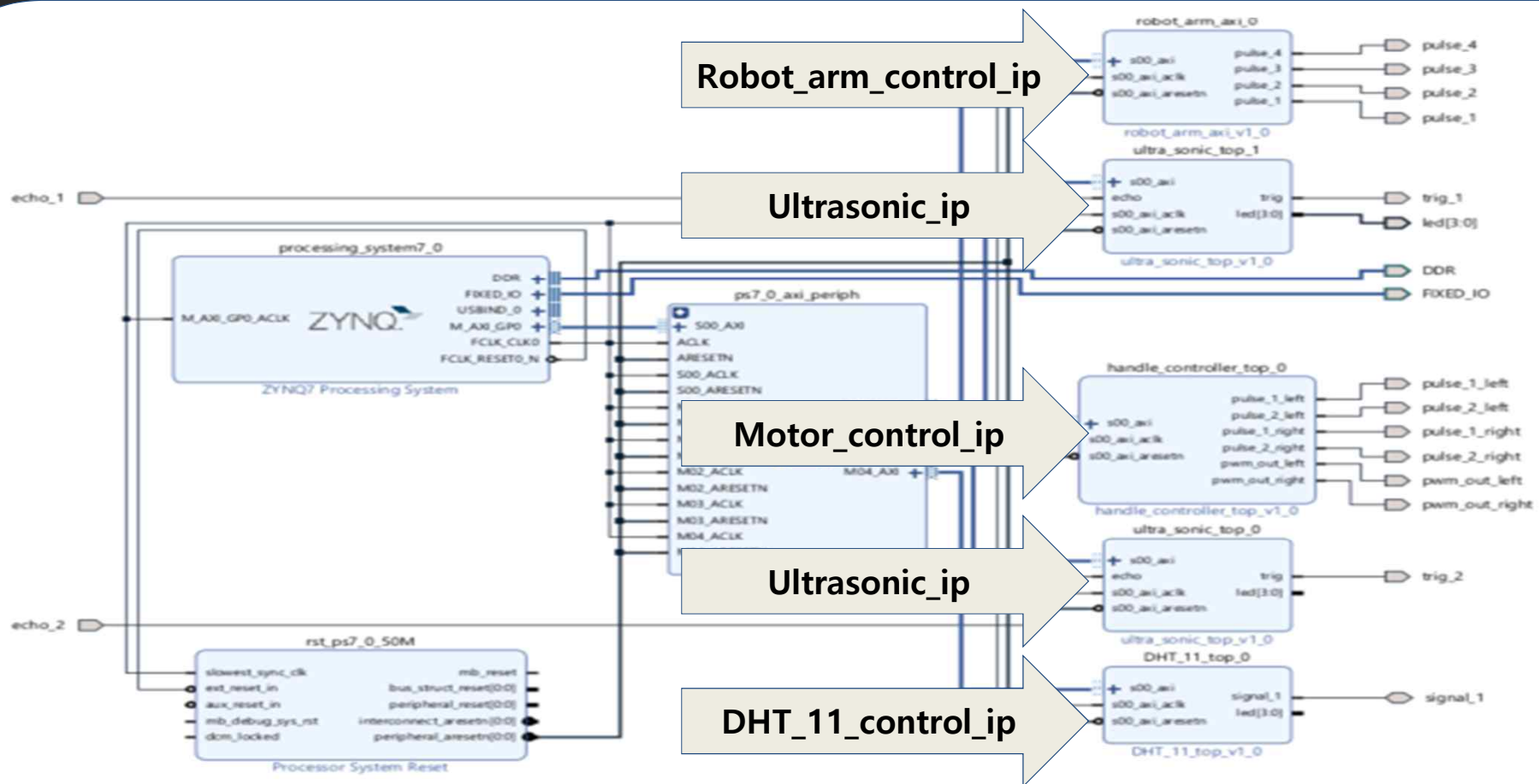


FPGA 구현

Ip block



IP Block



Raspberrypi 구현

UART 통신
OpenCV
C#



Raspberry Pi Code : UART 통신

```
import socket
from _thread import *
import serial
import time

fpga_uart = serial.Serial('/dev/ttyUSB1', baudrate=115200, timeout=1) # 시리얼 포트와 설정에 맞게 수정

velocity = 0
goback = 0
leftright = 0
now_area = 0

def readFPGAResponse():
    while True:
        try:
            response = fpga_uart.readline() # FPGA에서 데이터를 읽어옴
            if response:
                print('Received data from FPGA:', response.decode().strip())
        except serial.SerialException as e:
            print('Error reading data from FPGA:', str(e))
            break

def sendZeroToFPGA():
    while True:
        fpga_uart.write(b'00000\n') # FPGA로 0을 전송
        time.sleep(1) # 1초 대기

def sendData(operation, mode):
    if operation==1:
        if mode=='area1':
            fpga_uart.write(b'20\n')
            fpga_uart.write(b'00000\n')
        elif mode=='area2':
            fpga_uart.write(b'20\n')
            fpga_uart.write(b'00001\n')

    if not readFPGA:
        start_new_thread(readFPGAResponse, ())
        readFPGA = 1

    if not sendZero:
        start_new_thread(sendZeroToFPGA, ())
        sendZero = 1
```

UART

UART receive from FPGA

UART send to FPGA

Multi-thread

Debugging process : UART 통신

```
server start  
wait  
PC connected  
wait  
Connected by : 192.168.0.109 : 49237
```

```
velocity 0  
goback 0  
leftright 0  
now_area 1
```

```
Received data from FPGA: stopplz enter value  
Received data from FPGA: motor_1data == 1plz enter value
```

```
velocity 1  
goback 0  
leftright 0  
now_area 0  
Received data from FPGA: stopplz enter value  
Received data from FPGA: str_data == 0plz enter value  
Received data from FPGA: speed2data == 1plz enter value
```

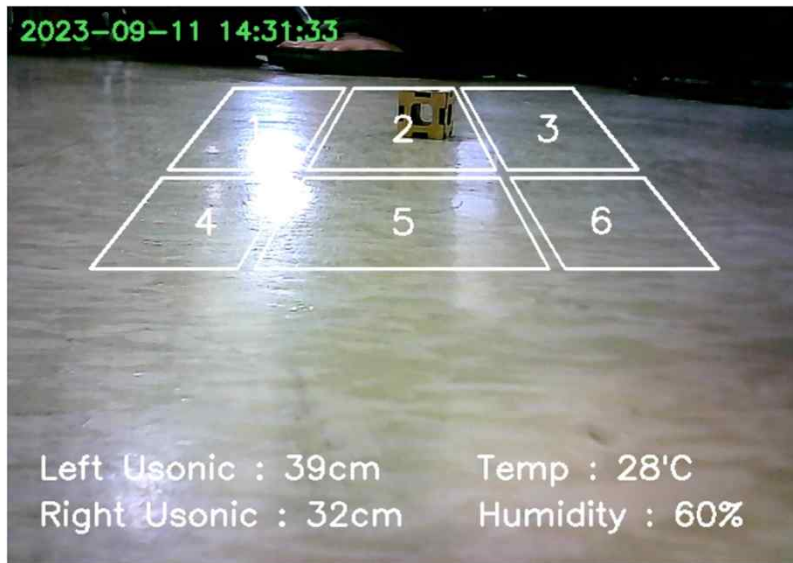
```
velocity 0  
goback 0  
leftright 0
```

PC에서 송신한 값 수신

FPGA에서 수신한 값 수신

Window Form / Website

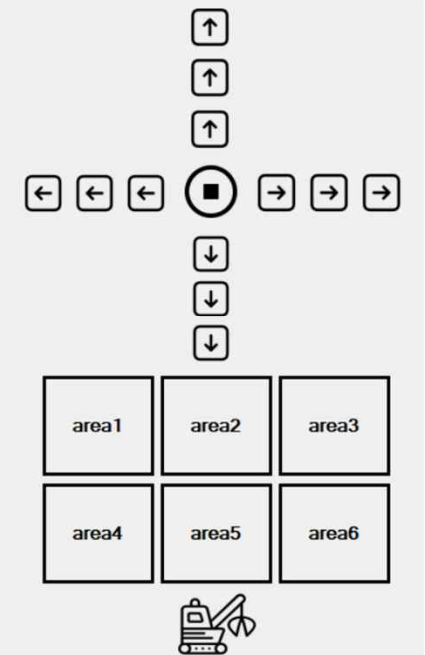
Tori Tori



Website

frm_Commclient

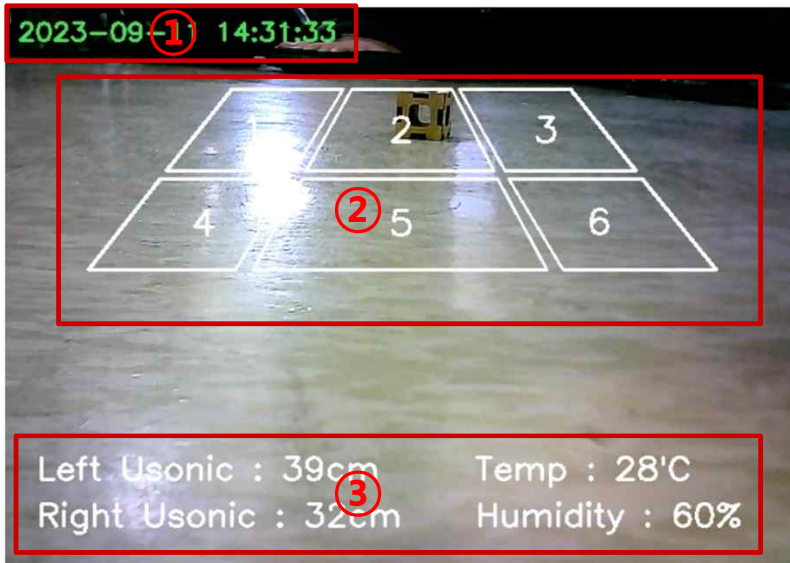
Tori Tori



Window Form

OpenCV

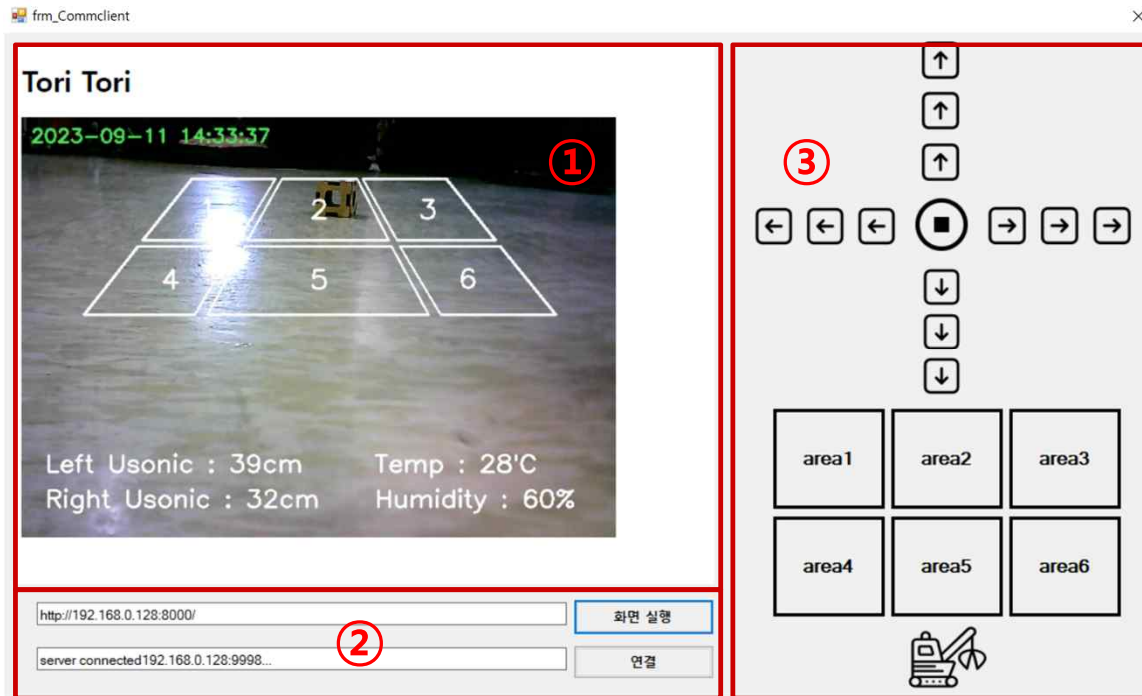
Tori Tori



- ① `putText (datetime)`
→ 현재 시간 표시
- ② `generate_frames`
→ AREA 표시할 다각형 생성
- ③ `putText`
→ 초음파센서, 온도 및 습도 표시

Website

C#



① url을 통해 영상출력

② RaspberryPi < > PC 간의 서버 클라이언트 연결

③ 버튼제어를 통해 ZYNQ의 센서 모듈 제어

Window Form

OpenCV code

```
95     #Clock
96     current_time = datetime.datetime.now().strftime('%Y-%m-%d %H:%M:%S')
97     position4 = (10, 28) # 텍스트를 추가할 위치 (x, y)
98     font = cv2.FONT_HERSHEY_SIMPLEX
99     font_scale = 0.7
100     font_color = (88, 220, 73) # BGR 형식의 색상 (녹색)
101     font_thickness = 2
102
103     cv2.putText(frame_with_polygon, current_time, position4, font, font_scale, font_color, font_thickness)
104
105 @app.route('/')
106 def index():
107     return render_template('index.html') # You need to create an HTML template
108
109 @app.route('/video_feed')
110 def video_feed():
111     return Response(generate_frames(), mimetype='multipart/x-mixed-replace; boundary=frame')
112
113 if __name__ == '__main__':
114     app.run(host='192.168.0.128', port=8000)
```

온도, 습도, 시계 등을 표시

URL 경로에 따라 요청을
처리할 수 있도록 라우팅

서버와 통신할 IP 및 PORT

Camera_Flask.py

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Tori Tori</title>
5 </head>
6 <body>
7     <h1>Tori Tori</h1>
8     
9 </body>
10 </html>
```

웹 구성 코드

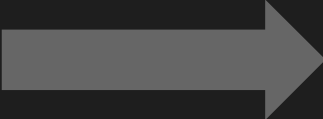
index.html

C# Code

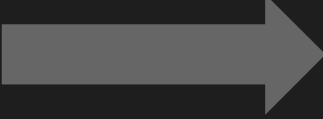
```
await webView.CoreWebView2.AddScriptToExecuteOnDocumentCreatedAsync("window.chrome.webview.postMessage(window.document.URL);");
await webView.CoreWebView2.AddScriptToExecuteOnDocumentCreatedAsync("window.chrome.webview.addEventListener('message', event => alert(event.data));");
}
참조 1개
void UpdateAddressBar(object sender, CoreWebView2WebMessageReceivedEventArgs args)
{
    String uri = args.TryGetWebMessageAsString();
    addressBar.Text = uri;
    webView.CoreWebView2.PostWebMessageAsString(uri);
}
참조 1개
private void Form_Resize(object sender, EventArgs e)
{
    webView.Size = this.ClientSize - new System.Drawing.Size(webView.Location);
    goButton.Left = this.ClientSize.Width - goButton.Width;
    addressBar.Width = goButton.Left - addressBar.Left;
}
참조 1개
private void goButton_Click(object sender, EventArgs e)
{
    if (webView != null && webView.CoreWebView2 != null)
    {
        webView.CoreWebView2.Navigate(addressBar.Text);
    }
}
try
{
    sock = new Socket(AddressFamily.InterNetwork, SocketType.Stream, ProtocolType.Tcp);
    sock.Connect("192.168.0.128", 9998); // ip, port 정보
    sock.Send(Encoding.Default.GetBytes("PC"));
    tb_client.Text += $"server connected(sock.RemoteEndPoint.ToString())...\n";
    ReadThread = new Thread(ReadProcess);
    ReadThread.Start();
}

private void SendTextToServer(string text)
{
    byte[] data = Encoding.Default.GetBytes(text); // 문자열을 바이트 배열로 변환
    sock.Send(data); // 서버로 데이터 전송
}

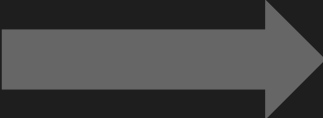
참조 1개
private void btn_area1_Click(object sender, EventArgs e)
{
    SendTextToServer("state area1");
}
```



영상 URL

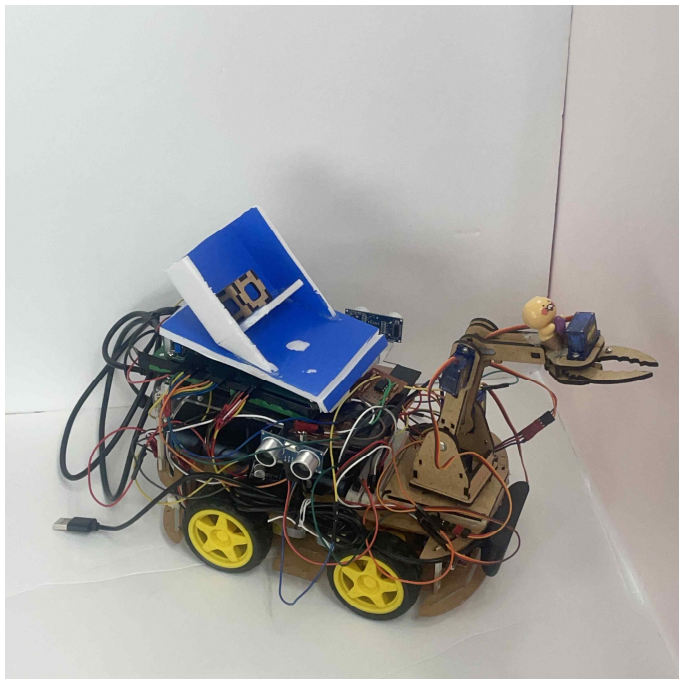


서버 접속



서버로 데이터 전송

결과물 사진 & 시연 영상



결과물



시연 영상

시행착오 & 개선사항

- ① WebCAM 영상이 포함된 웹 어플리케이션을 구축하기 위해 fastAPI를 사용하였지만 라즈베리파이와 WebCAM의 연동오류 발생
- ② 처음에 라즈베리파이에서 FPGA로 데이터를 전송할 때에 값이 제대로 전송되지 않음
- ③ DC-DC Converter를 사용하여 신호 증감 하려 했는데 신호가 시정수에 의해 천천히 올라가고 내려가는 현상이 발생
- ④ 초음파 센서 회로에서 setup time violation이 발생되었음.

시행착오



- ① fastAPI 웹 프레임워크 대신 Flask 웹 프레임워크를 사용함
- ② 데이터를 2진수로 변환시켜 보냄으로서 원하는 데이터를 전송할 수 있었음
- ③ lm358 op-amp를 이용하여 회로를 만들어 연결하여 신호 증감 해결함
- ④ 거리 변환 회로에서 나눗셈 연산이 아닌 fixed point 연산을 통해 Propagation delay를 줄임

개선사항



Q&A